Abdulrahman Sirelkhatim

Medical Graduate · Health Data Scientist · Organizational Leader

Omdurman, Sudan · +249 908 115 725 · abdulrahmanalsir4@gmail.com · abdoalsir.github.io · github.com/abdoalsir

Summary

First-generation medical graduate (MBBS, SUST, 2026) and health data scientist. Delivered 25 data analysis projects for medical researchers across 10+ specialties, co-authored studies on mass casualty preparedness in Sudan, and led SUMSA (800+ members) as President, all while completing medical school through revolution, COVID-19, a faculty strike and armed conflict. Targeting health data science research and graduate study in computational neuroscience.

Education

MBBS, Sudan University of Science and Technology · *11/2018 – 07/2026 · GPA 3.24/4.0 · Ranked 22nd of 127 · First Degree Honors · Thesis: Grade A*

Certificate in Computer & Data Science, MIT Emerging Talent Program · *11/2024 – 12/2025 · Capstone: HCICS geospatial health platform (Python, GeoPandas, Streamlit)*

Research

- **Trauma Surgeons' Preparedness for Mass Casualty Incidents in Sudan (2025)** and **Doctors' Perspectives of Hospital Preparedness**: co-author, two multicenter qualitative studies across 5 tertiary hospitals
- **Stroke KAP study** (MBBS thesis, Grade A): author; cross-sectional survey of 473 adults, analysed in Python and SPSS
- **Um Katti Hospital post-conflict baseline assessment**: main author of the final report; 262-household survey, 26 key informant interviews
- **Brain tumour classification on African MRI** (Grad-CAM, offline mobile deployment): main author, ongoing independent research

Experience

Freelance Health Data Analyst — *Self-employed* · *10/2025 – Present*

- Delivered 25 analysis projects for MD, MBBS, Master's and doctoral researchers: study design, regression, survival and ROC analysis, figures, and thesis-defense presentations (11 prepared, all clients defended)

Deputy Director, Resources & Excellence Management — *Sudanese Consortium for Surgical Development* · *08/2024 – 05/2026*

- Led recruitment across 4 departments (80+ applications, 20–30 interviews); defined KPIs for every position; built the M&E framework

Organizer, Fundamentals of Nursing Course — *Sharg Alneel Hospital* · *10/2026 – 12/2026*

- Delivered 3 sessions to nursing staff; created a facilitator portfolio of 43 sessions and 106 hours across 9 universities

Leadership

President, Sudan University Medical Students Association (SUMSA) · *03/2025 – 12/2025 · 800+ students*

- Rebuilt the association from near-collapse within 2 months; designed its M&E system; led constitutional reform ratified by 400+ voters; ran a 7-day orientation for 465 students

Secretary General (de facto President), SUMSA · *2023*

- Built a 14,000+ document archive; raised 12,000,000+ SDG supporting 98+ war-affected students

IFMSA / MedSIN-Sudan (SCOPE) · *2020 – 2024*

- Founded the TNEF facilitator training (certified facilitators 7 to 20+); IFMSA Certified Facilitator (PRET, 2022)

Skills

Data: Python (pandas, scikit-learn, statsmodels, GeoPandas), SPSS, SQL, Excel, Git/GitHub
Methods: Regression, survival analysis, diagnostic accuracy, survey design, machine learning, Grad-CAM
Clinical: Rotations across 10 specialties; thesis and field research

Certifications & Languages

IBM Data Science Professional · DeepLearning.AI Math for ML · Google Project Management · MITx 6.00.1x/2x · IFMSA Certified Facilitator · IFMSA OGHA Workshop
Arabic (native) · English (fluent, C1, Duolingo 140)